THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(d) $\frac{1}{2}$	Favourable outcomes: 2, 4, 6; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
2	(d) $\frac{1}{3}$	Favourable outcomes: 3, 6; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
3	(d) $\frac{1}{4}$	Favourable outcomes: HH; number $= 1$; $P = \frac{1}{4}$
4	(b) $\frac{3}{8}$	Favourable balls $= 3$; $P = \frac{3}{8}$
5	(c) $\frac{3}{10}$	Favourable numbers: 3, 6, 9; number $= 3$; $P = \frac{3}{10}$
6	(c) $\frac{3}{4}$	$P(\text{not } E) = 1 - \frac{1}{4}$; $P(\text{not } E) = \frac{3}{4}$
7	(b) 1	A sure event always happens
8	(b) $\frac{1}{2}$	Favourable outcomes: 1, 3, 5; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
9	(c) $\frac{1}{3}$	Favourable outcomes: 1, 4; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
10	(c) $\frac{3}{4}$	Favourable outcomes: HH, HT, TH; number $= 3$; $P = \frac{3}{4}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total possible outcomes $= 6$: 1, 2, 3, 4, 5, 6; Favourable outcomes: ; number $= 0$ (1); $P = \frac{0}{6} = 0$ (1).	1+1
12	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: HH; number $= 1$ (1); $P = \frac{1}{4}$ (1).	1+1
13	Total balls $= 5 + 3 = 8$; Favourable balls $= 3$ (1); $P = \frac{3}{8}$ (1).	1+1
14	Total possible outcomes $= 6$: 1, 2, 3, 4, 5, 6; Favourable outcomes: 5, 6; number $= 2$ (1); $P = \frac{2}{6} = \frac{1}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total tickets $= 10 + 25 = 35$ (1); Favourable tickets (prizes) $= 10$ (1); $P = \frac{10}{35} = \frac{2}{7}$ (1).	1+1+1
16	Total pens $= 144$; good pens $= 144 - 20 = 124$ (1); Favourable outcomes $=$ number of good pens (1); $P = \frac{124}{144} = \frac{31}{36}$ (1).	1+1+1
17	Total students $= 40$; boys $= 40 - 24 = 16$ (1); Favourable outcomes $= 16$ (1); $P = \frac{16}{40} = \frac{2}{5}$ (1).	1+1+1
	OR: Total tickets $= 8 + 32 = 40$ (1); Favourable tickets (prizes) $= 8$ (1); $P = \frac{8}{40} = \frac{1}{5}$ (1).	1+1+1
18	Total pens $= 100$; good pens $= 100 - 12 = 88$ (1); Favourable outcomes $=$ number of good pens (1); $P = \frac{88}{100} = \frac{22}{25}$ (1).	1+1+1
	OR: Total students $= 30$; boys $= 30 - 12 = 18$ (1); Favourable outcomes $= 18$ (1); $P = \frac{18}{30} = \frac{3}{5}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total balls $= 10$ (2); Favourable balls: 3, 6, 0 (2); Probabilities: $\frac{3}{10}$, $\frac{3}{5}$, 0 (1).	2+2+1
	OR: Total outcomes $= 12$ (2); Favourable counts: 9, 5, 0 (2); Probabilities: $\frac{3}{4}$, $\frac{5}{12}$, 0 (1).	2+2+1
20	Total balls $= 20$ (2); Favourable balls: 8, 12, 0 (2); Probabilities: $\frac{2}{5}$, $\frac{3}{5}$, 0 (1).	2+2+1
	OR: Total outcomes $= 20$ (2); Favourable counts: 6, 8, 5 (2); Probabilities: $\frac{3}{10}$, $\frac{2}{5}$, $\frac{1}{4}$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(d) $\frac{3}{5}$	Favourable balls $= 6$; $P = \frac{6}{10} = \frac{3}{5}$
2	(a) $\frac{5}{12}$	Favourable numbers: 2, 3, 5, 7, 11; number $= 5$; $P = \frac{5}{12}$
3	(d) $\frac{4}{7}$	$P(\text{not } E) = 1 - \frac{3}{7}$; $P(\text{not } E) = \frac{4}{7}$
4	(d) 0	It never happens
5	(c) $\frac{1}{3}$	Favourable outcomes: 1, 2; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
6	(a) $\frac{2}{3}$	Favourable outcomes: 1, 2, 3, 6; number $= 4$; $P = \frac{4}{6} = \frac{2}{3}$
7	(d) $\frac{1}{4}$	Favourable outcomes: TT; number $= 1$; $P = \frac{1}{4}$
8	(a) $\frac{1}{5}$	Favourable balls $= 2$; $P = \frac{2}{10} = \frac{1}{5}$
9	(a) $\frac{2}{15}$	Favourable numbers: 6, 12; number $= 2$; $P = \frac{2}{15}$
10	(c) $\frac{7}{9}$	$P(\text{not } E) = 1 - \frac{2}{9}$; $P(\text{not } E) = \frac{7}{9}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: HH, HT, TH; number $= 3$ (1); $P = \frac{3}{4}$ (1).	1+1
12	Total balls $= 2 + 6 = 8$; Favourable balls $= 2$ (1); $P = \frac{2}{8} = \frac{1}{4}$ (1).	1+1
13	Total possible outcomes $= 6$: 1, 2, 3, 4, 5, 6; Favourable outcomes: 4, 6; number $= 2$ (1); $P = \frac{2}{6} = \frac{1}{3}$ (1).	1+1
14	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: TT; number $= 1$ (1); $P = \frac{1}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total tickets $= 5 + 45 = 50$ (1); Favourable tickets (prizes) $= 5$ (1); $P = \frac{5}{50} = \frac{1}{10}$ (1).	1+1+1
16	Total pens $= 50$; good pens $= 50 - 5 = 45$ (1); Favourable outcomes = number of good pens (1); $P = \frac{45}{50} = \frac{9}{10}$ (1).	1+1+1
17	Total students $= 50$; boys $= 50 - 28 = 22$ (1); Favourable outcomes $= 22$ (1); $P = \frac{22}{50} = \frac{11}{25}$ (1).	1+1+1
	OR: Total tickets $= 12 + 28 = 40$ (1); Favourable tickets (prizes) $= 12$ (1); $P = \frac{12}{40} = \frac{3}{10}$ (1).	1+1+1
18	Total pens $= 80$; good pens $= 80 - 16 = 64$ (1); Favourable outcomes = number of good pens (1); $P = \frac{64}{80} = \frac{4}{5}$ (1).	1+1+1
	OR: Total students $= 36$; boys $= 36 - 20 = 16$ (1); Favourable outcomes $= 16$ (1); $P = \frac{16}{36} = \frac{4}{9}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total balls $= 10$ (2); Favourable balls: 3, 8, 7 (2); Probabilities: $\frac{3}{10}$, $\frac{4}{5}$, $\frac{7}{10}$ (1).	2+2+1
	OR: Total outcomes $= 15$ (2); Favourable counts: 2, 3, 15 (2); Probabilities: $\frac{2}{15}$, $\frac{1}{5}$, 1 (1).	2+2+1
20	Total balls $= 20$ (2); Favourable balls: 6, 10, 0 (2); Probabilities: $\frac{3}{10}$, $\frac{1}{2}$, 0 (1).	2+2+1
	OR: Total outcomes $= 18$ (2); Favourable counts: 3, 4, 9 (2); Probabilities: $\frac{1}{6}$, $\frac{2}{9}$, $\frac{1}{2}$ (1).	2+2+1

Mistakes to avoid:

- Writing a probability greater than 1 or negative; a probability always lies between 0 and 1.
- Using outcomes that are not equally likely (e.g. treating 'sum 7' and 'sum 12' as equally likely).
- Counting 1 as a prime number, or leaving out 2, the only even prime.
- Counting the overlap twice in 'A or B' questions.
- For two dice, using only 12 or 6 outcomes instead of 36, or counting (2,6) and (6,2) as the same outcome.
- Not changing the total number of outcomes when balls are removed or added, and not simplifying the fraction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Probability** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(d) $\frac{7}{6}$	Probability lies between 0 and 1
2	(b) $\frac{1}{3}$	Favourable outcomes: 5, 6; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
3	(a) $\frac{1}{3}$	Favourable outcomes: 3, 4; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
4	(c) $\frac{1}{2}$	Favourable outcomes: HT, TH; number $= 2$; $P = \frac{2}{4} = \frac{1}{2}$
5	(a) $\frac{3}{10}$	Favourable balls $= 3$; $P = \frac{3}{10}$
6	(b) $\frac{1}{4}$	Favourable numbers: 4, 8, 12, 16, 20; number $= 5$; $P = \frac{5}{20} = \frac{1}{4}$
7	(b) $\frac{7}{12}$	$P(\text{not } E) = 1 - \frac{5}{12}$; $P(\text{not } E) = \frac{7}{12}$
8	(b) -0.3	Probability is never negative
9	(b) $\frac{1}{2}$	Favourable outcomes: 2, 3, 5; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
10	(a) 1	Favourable outcomes: 1, 2, 3, 4, 5, 6; number $= 6$; $P = \frac{6}{6} = 1$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total balls $= 4 + 6 = 10$; Favourable balls $= 6$ (1); $P = \frac{6}{10} = \frac{3}{5}$ (1).	1+1
12	Total possible outcomes $= 6$: 1, 2, 3, 4, 5, 6; Favourable outcomes: 2; number $= 1$ (1); $P = \frac{1}{6}$ (1).	1+1
13	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: HT, TH; number $= 2$ (1); $P = \frac{2}{4} = \frac{1}{2}$ (1).	1+1
14	Total balls $= 8 + 2 = 10$; Favourable balls $= 2$ (1); $P = \frac{2}{10} = \frac{1}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total tickets $= 6 + 14 = 20$ (1); Favourable tickets (prizes) $= 6$ (1); $P = \frac{6}{20} = \frac{3}{10}$ (1).	1+1+1
16	Total pens $= 120$; good pens $= 120 - 18 = 102$ (1); Favourable outcomes = number of good pens (1); $P = \frac{102}{120} = \frac{17}{20}$ (1).	1+1+1
17	Total students $= 45$; boys $= 45 - 18 = 27$ (1); Favourable outcomes $= 27$ (1); $P = \frac{27}{45} = \frac{3}{5}$ (1).	1+1+1
	OR: Total tickets $= 9 + 21 = 30$ (1); Favourable tickets (prizes) $= 9$ (1); $P = \frac{9}{30} = \frac{3}{10}$ (1).	1+1+1
18	Total pens $= 60$; good pens $= 60 - 9 = 51$ (1); Favourable outcomes = number of good pens (1); $P = \frac{51}{60} = \frac{17}{20}$ (1).	1+1+1
	OR: Total students $= 25$; boys $= 25 - 10 = 15$ (1); Favourable outcomes $= 15$ (1); $P = \frac{15}{25} = \frac{3}{5}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total balls $= 20$ (2); Favourable balls: 6, 5, 14 (2); Probabilities: $\frac{3}{10}$, $\frac{1}{4}$, $\frac{7}{10}$ (1).	2+2+1
	OR: Total outcomes $= 25$ (2); Favourable counts: 5, 5, 0 (2); Probabilities: $\frac{1}{5}$, $\frac{1}{5}$, 0 (1).	2+2+1
20	Total balls $= 20$ (2); Favourable balls: 16, 8, 20 (2); Probabilities: $\frac{4}{5}$, $\frac{2}{5}$, 1 (1).	2+2+1
	OR: Total outcomes $= 30$ (2); Favourable counts: 5, 6, 10 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{5}$, $\frac{1}{3}$ (1).	2+2+1

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- Counting 1 as a prime number, or leaving out 2, the only even prime.
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